

Fundamentals of Information Systems Security

Linux Commands for Security Professionals

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Quick Reference Guide

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1 File System Commands

Command	Syntax	Description
pwd	pwd	Print working directory
ls	ls -la	List files with all details
cd	cd /path	Change directory
mkdir	mkdir dirname	Create directory
rmdir	rmdir dirname	Remove empty directory
rm	rm -rf file	Remove file/directory recursively
cp	cp -r source dest	Copy files/directories
mv	mv source dest	Move or rename files
find	find / -name "*.txt"	Find files by name
locate	locate filename	Quickly find files (updatedb first)
which	which command	Show path of command
whereis	whereis command	Show binary, source, and man pages
chmod	chmod 755 file	Change file permissions
chown	chown user:group file	Change file ownership
ln	ln -s target link	Create symbolic link

2 User and Permission Management

Command	Syntax	Description
whoami	whoami	Show current username
id	id username	Show user ID and group info
who	who	Show logged-in users
w	w	Show system uptime and users
last	last	Show last logins
useradd	sudo useradd -m username	Add new user
usermod	sudo usermod -aG group user	Modify user (add to group)
userdel	sudo userdel -r username	Delete user
passwd	passwd	Change password
groupadd	sudo groupadd groupname	Add new group
groupdel	sudo groupdel groupname	Delete group
su	su - username	Switch user
sudo	sudo command	Execute as root

3 Process Management

Command	Syntax	Description
ps	ps aux	Show all running processes
ps	ps auxf	Show tree of processes
top	top	Interactive process viewer
htop	htop	Enhanced process viewer
kill	kill -9 PID	Force kill process
killall	killall processname	Kill all process by name
pkill	pkill -f pattern	Kill process by pattern
jobs	jobs	List background jobs
bg	bg %job	Resume job in background
fg	fg %job	Bring job to foreground
nice	nice -n 10 command	Run command with priority
renice	renice 10 -p PID	Change process priority

4 Network Commands

Command	Syntax	Description
ifconfig	ifconfig -a	Show all network interfaces
ip	ip addr show	Show IP addresses (modern)
ip	ip route show	Show routing table
ping	ping -c 4 host	Test connectivity
traceroute	traceroute host	Trace network path
netstat	netstat -tulpn	Show listening ports
ss	ss -tulpn	Show socket statistics (modern)
nmap	nmap -sS target	SYN scan (needs root)
nmap	nmap -sV target	Version detection scan
tcpdump	tcpdump -i eth0	Capture network traffic
wireshark	wireshark	GUI packet analyzer
curl	curl -I URL	Get HTTP headers
wget	wget URL	Download file
nc	nc -lvp 4444	Listen on port (Netcat)
nc	nc host 4444	Connect to host (Netcat)
arp	arp -a	Show ARP cache
route	route -n	Show routing table

5 System Information

Command	Syntax	Description
uname	uname -a	Show system info
hostname	hostname	Show hostname
uptime	uptime	Show uptime
free	free -h	Show memory usage
df	df -h	Show disk usage
du	du -sh directory	Show directory size
lscpu	lscpu	Show CPU info
lsblk	lsblk	Show block devices
dmesg	dmesg tail	Show kernel messages

6 Log Analysis Commands

Command	Syntax	Description
cat	cat /var/log/syslog	Display file content
less	less /var/log/syslog	View file interactively
tail	tail -f /var/log/syslog	Follow log in real-time
head	head -20 /var/log/syslog	Show first 20 lines
grep	grep "error" logfile	Search in file
grep	grep -R "error" /var/log/	Recursively search
awk	awk '/pattern/ print \$1'	Print first column
sed	sed 's/old/new/g' file	Replace text
wc	wc -l file	Count lines
sort	sort file	Sort lines
uniq	uniq -c	Count unique lines

7 Security and Forensic Commands

Command	Syntax	Description
john	john --wordlist=file hash.txt	Password cracking
hashcat	hashcat -m 0 hash.txt wordlist	GPU password cracking
aircrack	aircrack-ng capture.cap	Crack WEP/WPA
airmon	airmon-ng start wlan0	Enable monitor mode

airodump	airodump-ng wlan0mon	Capture wireless packets
metasploit	msfconsole	Start Metasploit
msfvenom	msfvenom -p payload	Generate exploit payloads
openssl	openssl enc -aes-256-cbc	Encrypt/decrypt files
gpg	gpg -c file	Encrypt file with GPG
ssh-keygen	ssh-keygen -t rsa	Generate SSH key
dd	dd if=/dev/sda of=image.dd	Create disk image
strings	strings file	Extract printable strings
xxd	xxd file	Hex dump file
binwalk	binwalk file	Analyze binary file
foremost	foremost -i image.dd	Recover files
scalpel	scalpel image.dd	File carving

8 Quick Command Reference

Top 20 Essential Commands

- **ls -la** - List all files
- **pwd** - Current directory
- **cd ~** - Home directory
- **sudo** - Run as root
- **apt update** - Update packages
- **apt install** - Install package
- **systemctl start** - Start service
- **systemctl stop** - Stop service
- **systemctl status** - Service status
- **journalctl -xe** - View system logs
- **ps aux** - All processes
- **top** - Process monitoring
- **netstat -tulpn** - Listening ports
- **ss -tulpn** - Socket stats
- **ping host** - Test connectivity
- **traceroute host** - Network trace
- **tcpdump -i eth0** - Capture traffic
- **grep pattern file** - Search
- **find / -name file** - Find file
- **chmod 755 file** - Change permissions

9 File Permissions Reference

Permission Numeric Values

Value	Permission	Symbolic	Description
0	—	No Access	No permissions
1	-x	Execute	Can execute file
2	-w-	Write	Can write to file
3	-wx	Write + Execute	Can write and execute
4	r-	Read	Can read file
5	r-x	Read + Execute	Can read and execute
6	rw-	Read + Write	Can read and write
7	rwX	Full	Full permissions

Common Permissions:

- `chmod 755 file - rwxr-xr-x` (Owner: rwx, Group: r-x, Others: r-x)
- `chmod 644 file - rw-r-r-` (Owner: rw, Group: r-, Others: r-)
- `chmod 600 file - rw-----` (Owner: rw, Group: —, Others: —)